

ANA 222 Pectoral Region

By Dr. Mbaka G.O.

- A name given to a part situated in front of chest on either side of sternum.
- On it is the breast, superficial fascia, deep fascia, muscles, vessels, nerves and the rib cage.
- Superficial fascia: Lies under the skin and absent under the areola. It splits into two to enclose the mammary gland therefore mammary gland lies on it.
- It is an ill defined capsule with numerous septa running through the capsule dividing the mammary gland into compartments.
- Cutaneous nerves: Anterior intercostal pierces pectoral major lateral of sternum supplying adjacent skin. The nerves are accompanied by anterior cutaneous artery from internal mammary artery. Lateral cutaneous are branches of intercostal nerves.
- Pectoral fascia: A name to the deep fascia covering the pectoralis major muscle.
- Superiorly attaches to clavicle extending medially to sternum.

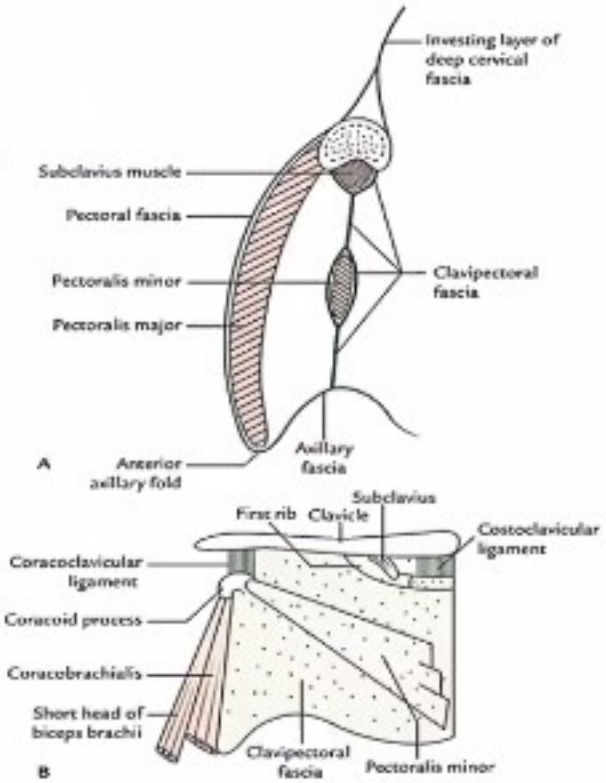
From inferior border of pectoralis major the fascia run posteriorly to enclose latissimus dorsi. Portion of the fascia between the muscles form deep fascia of the floor of axilla (axillary fascia). An upward extension of the fascia reaches the lower border of pectoralis minor (deep to pectoralis major) splitting to enclose the muscle.

At the upper border of the muscle the sheet unit forming clavipectoral fascia which runs upward to enclose the subclavius muscle and gets attached to the clavicle.

The fascia between the fascia of floor of axilla and the lower border of pectoralis minor is referred to as suspensory ligament of axilla.

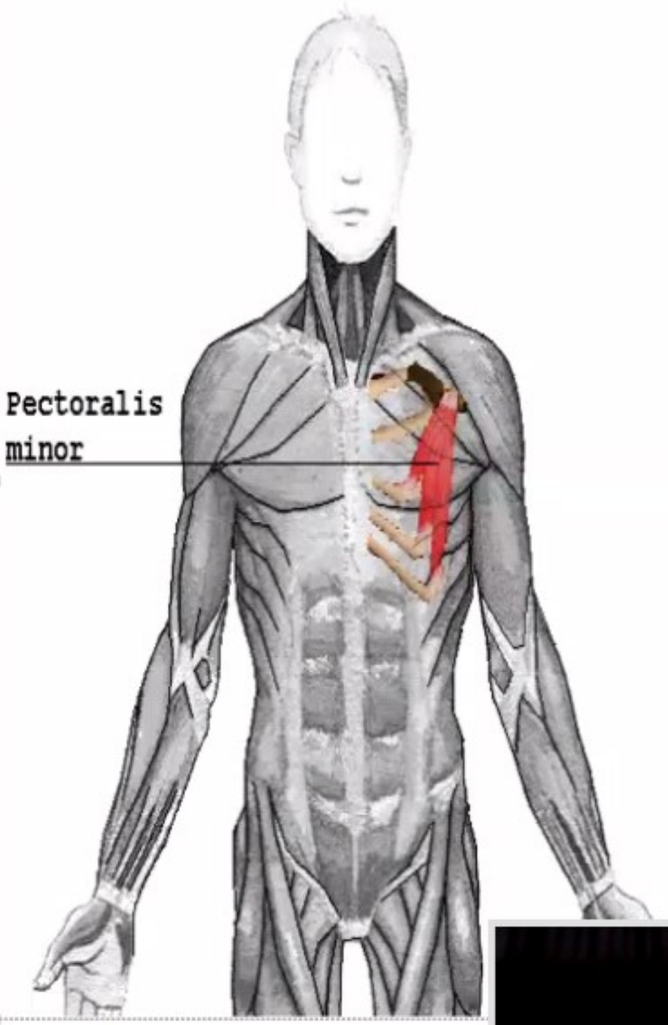
During abduction the clavicle is raised and the suspensory ligament is also pulled up thus making the hollow of the axilla more prominent.

Laterally the fascia continue with the deep fascia of deltoid while inferiorly with facia of external oblique muscle.

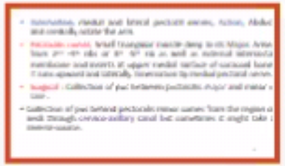


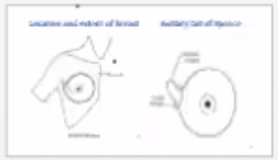
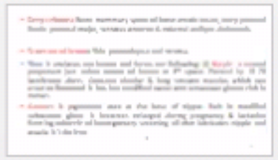
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- Pectoralis major: Largest muscle of pectoral region, triangular or fan shaped.
- Origin: Has two heads, clavicular and sternocostal; The cavicular rises from anterior surface of clavicle from medial 1/2 to medial 2/3 fibres run downward and lateral.
- The sternocostal fibres from front of manubrium and body of sternum reaching the level of 6th costal cartilage. Also from 2nd to 6th cartilage and occasionally from 1st. Small portion from external oblique aponeurosis. The two heads are separated by deltoid groove for cephalic vein.
- Insertion: It inserts via bilaminar tendon; Fibres from clavicular head run laterally downward superficial to the 2nd head attaching at lower part of lateral lip of bicipital groove while fibres from sternocostal head ascend upward and laterally deep to the tendon of clavicular head attaching at the upper part of lateral lip of bicipital groove.





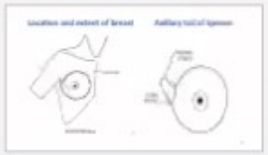


- Innervation: medial and lateral pectoral nerves; Action, Abduct and medially rotate the arm.
- Pectoralis minor: Small triangular muscle deep to its Major. Arises from 2nd -4th ribs or 3rd -5th rib as well as external intercostal membrane and inserts at upper medial surface of coracoid bone. It runs upward and laterally. Innervation by medial pectoral nerve.
- Surgical : Collection of pus between pectoralis major and minor is rare .
- Collection of pus behind pectoralis minor comes from the region of neck through cervico-axillary canal but sometimes it might take a reverse course.






The Breast

- A modified sweat gland present in both sexes.
- Develops in females after puberty and secrets milk. It's an accessory organ of female reproductive system.
- An apocrine gland situated on superficial fascia of pectoral region & small portion (axillary tail) pierces deep fascia to lie in axilla.
- In young adult female is spherical and later pendulous.
- Vertically extends from 2nd to 6th rib. Horizontally at level of 4th costal cartilage from lateral sternal border to midaxillary line.
- Superior laterally prolong laterally pierces the deep fascia, the known as foramen of Langer to lie at 3rd rib & that part is known as axillary tail of Spence.

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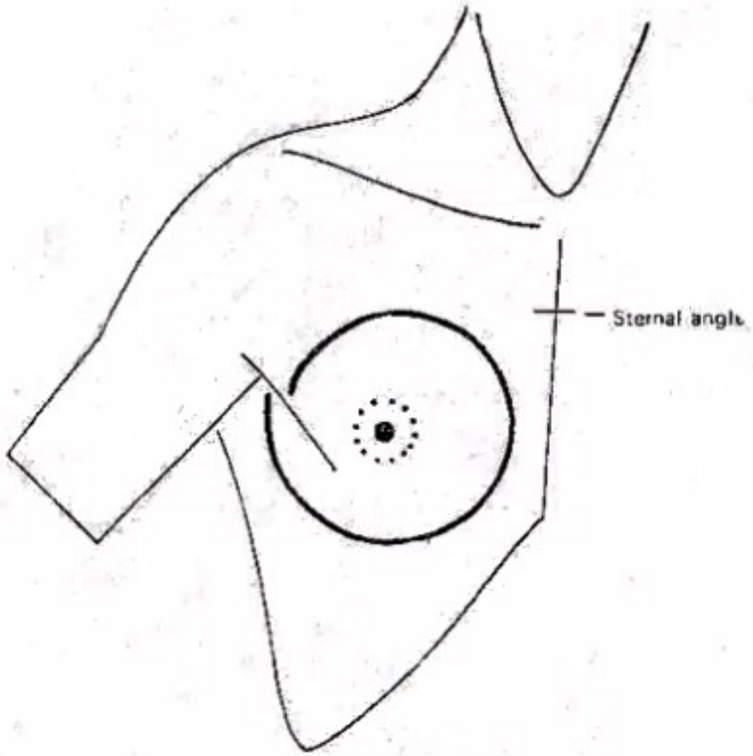
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5 The Breast

6 Location and extent of breast Axillary tail of Spence

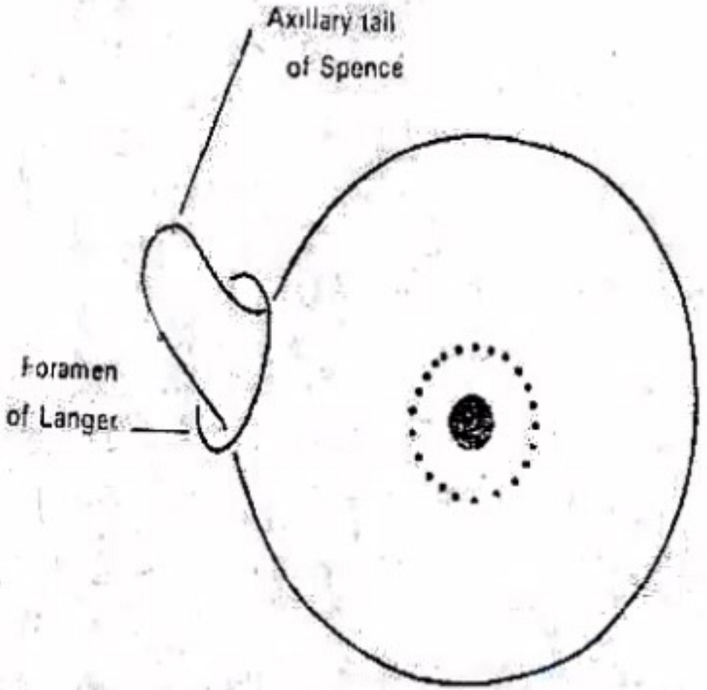
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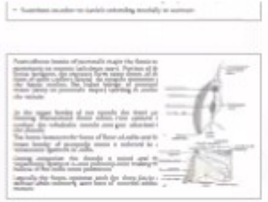
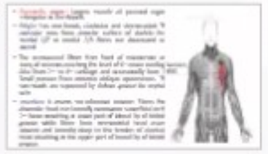
Location and extent of breast

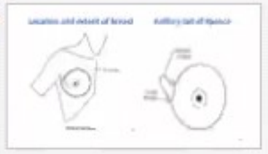


Extent of the breast.

Axillary tail of Spence




- Deep relation: Retro mammary space of loose areola tissue, deep pectoral fascia, pectoral major, serratus anterior & external oblique abdominis.

- Structure of breast: Skin, parenchyma and stroma.
- Skin: It encloses the breast and forms the following; (i) Nipple- a conical projection just below centre of breast at 4th space. Pierced by 15-20 lactiferous ducts. Contains circular & long smooth muscles which can erect or flattened. It has few modified sweat and sebaceous glands rich in nerves.
- Areolar: A pigmented area at the base of nipple. Rich in modified sebaceous gland. It becomes enlarged during pregnancy & lactation forming tubercle of Montgomery secreting oil that lubricates nipple and aroela. It's fat free.

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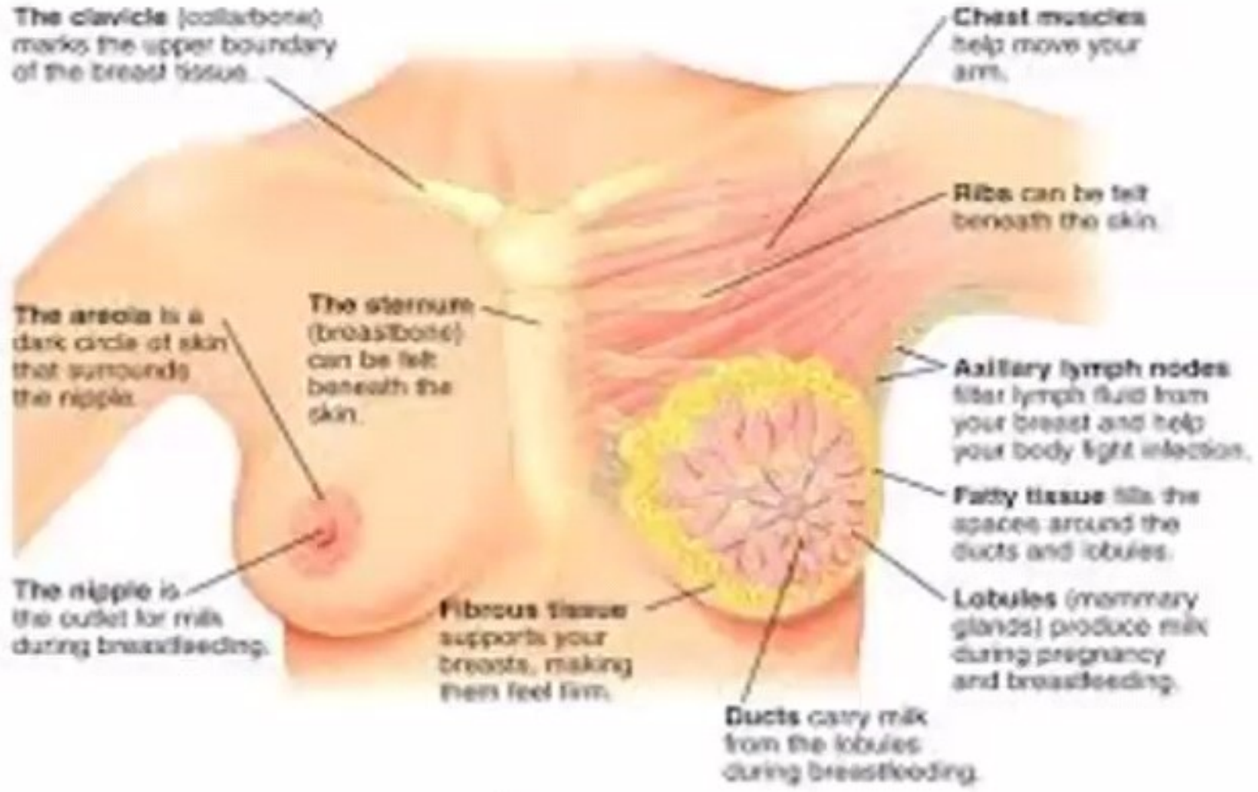
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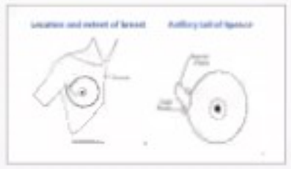
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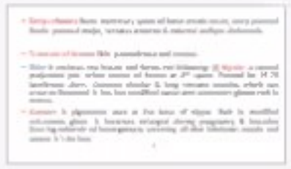
The nipple and areola



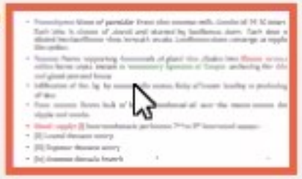












- Parenchyma: Made of glandular tissue that secretes milk. Consist of 15-20 lobes. Each lobe is cluster of alveoli and drained by lactiferous ducts. Each duct is dilated into lactiferous sinus beneath areola. Lactiferous ducts converge at nipple like spokes.
- Stroma: Forms supporting framework of gland that divides into fibrous stroma which forms septa known as suspensory ligament of Cooper anchoring the skin and gland pectoral fascia.
- Infiltration of this lig. by cancer cells causes fixity of breast leading to puckering of skin.
- Fatty stroma: Forms bulk of breast distributed all over the breast except the nipple and areala.
- Blood supply: (i) Internal thoracic perforates 2nd to 6th intercostal spaces.
- (ii) Lateral thoracic artery.
- (iii) Superior thoracic artery.
- (iv) Acromio-thoracic branch.

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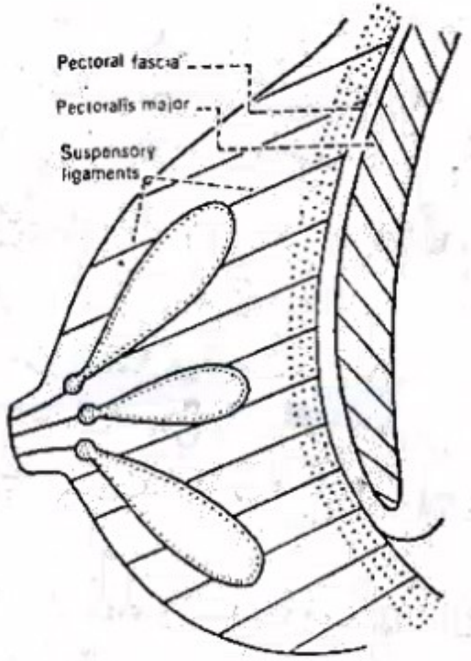
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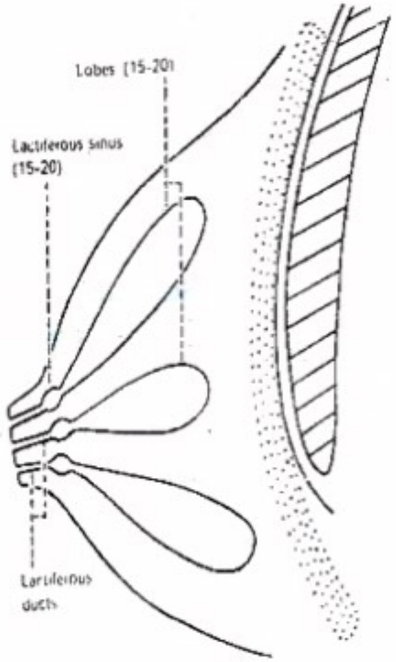
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Stromal fibres

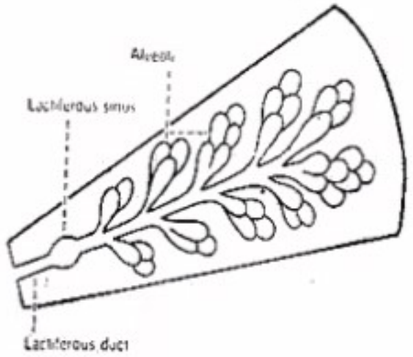


Suspensory ligaments of the breast.

Parenchyma of breast



Lobes of the mammary gland.



Structure of one lobe of the mammary gland.

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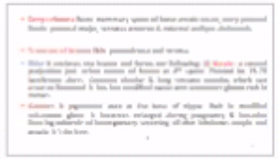
Clinical Anatomy

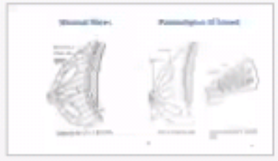
- Breast is the common site for tumour.
- The lymphatic channel is the pathway for carcinoma. Cancer cells may infect suspensory ligament making the breast fixed. Contraction can cause retraction or puckering of skin.
- Tumour in the breast can be benign or malignant.
- Benign tumour papiloma can be treated by surgical excision of the growth.
- Malignant tumour, breast cancer, is treated by the side affected or the two sides, a surgical procedure known as radical mastectomy.

Polymastia



- Extra breast most commonly occur along the milk line, usually just underneath the normally located breasts or nipples; however, they have also been noted in ectopic sites such as the back or the buttock.
- These accessory tissues may undergo malignant transformation and should be removed.





Lymphatic drainage of breast

- It's important in connection to disease spread.
- Divided into superficial and deep components with both freely communicating.
- Superficial nodes drain the skin of the breast except areola and nipples.
- Deep nodes drain the parenchyma, nipple and areola.
- The nodes are arranged into lateral upper and lower quadrant & medial upper and lower quadrant.
- Lateral part drain into axillary & infraclavicular nodes
- Medial part drain thru. Intercostal spaces to parasternal nodes.
- The two communicate due to pressure from clothing.
- Thru. Supraclavicular nodes gets to cephalic nodes, thru. Intercostal to posterior intercostal nodes, inferiorly to sub diaphragmatic to peritoneal nodes.
- 70% to axillary, 20% to mammary, 5% to post. Intercostal nodes.